

3. LOADING MECHANISM and OPERATION

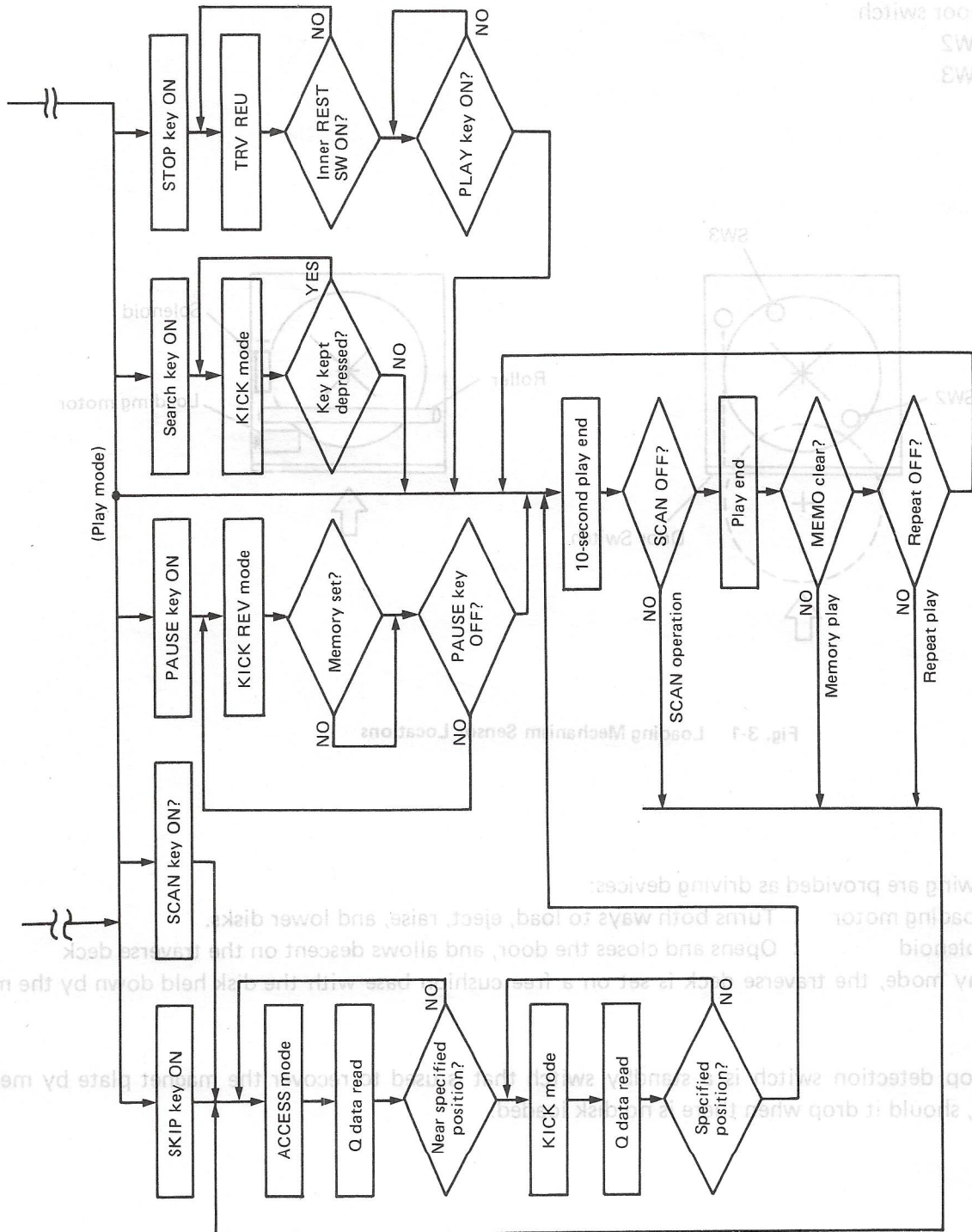
3-1 Disk Loading Mechanism

The following optical switches are provided as sensors to detect disk position.

Door switch

SW2

SW3



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- SW3

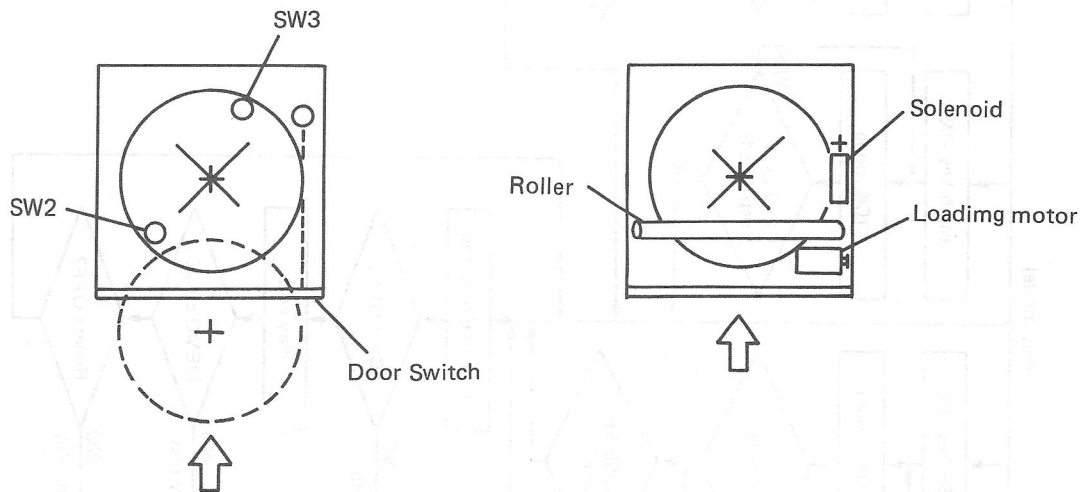


Fig. 3-1 Loading Mechanism Sensor Locations

The following are provided as driving devices:

- Loading motor : Turns both ways to load, eject, raise, and lower disks.
- Solenoid : Opens and closes the door, and allows descent on the traverse deck

In the play mode, the traverse deck is set on a free cushion base with the disk held down by the magnet plate.

* The drop detection switch is a standby switch that is used to recover the magnet plate by means of EJECT, should it drop when there is no disk loaded.

2. BLOCK DIAGRAM and CD SYSTEM OPERATION

2-1 Block Diagram

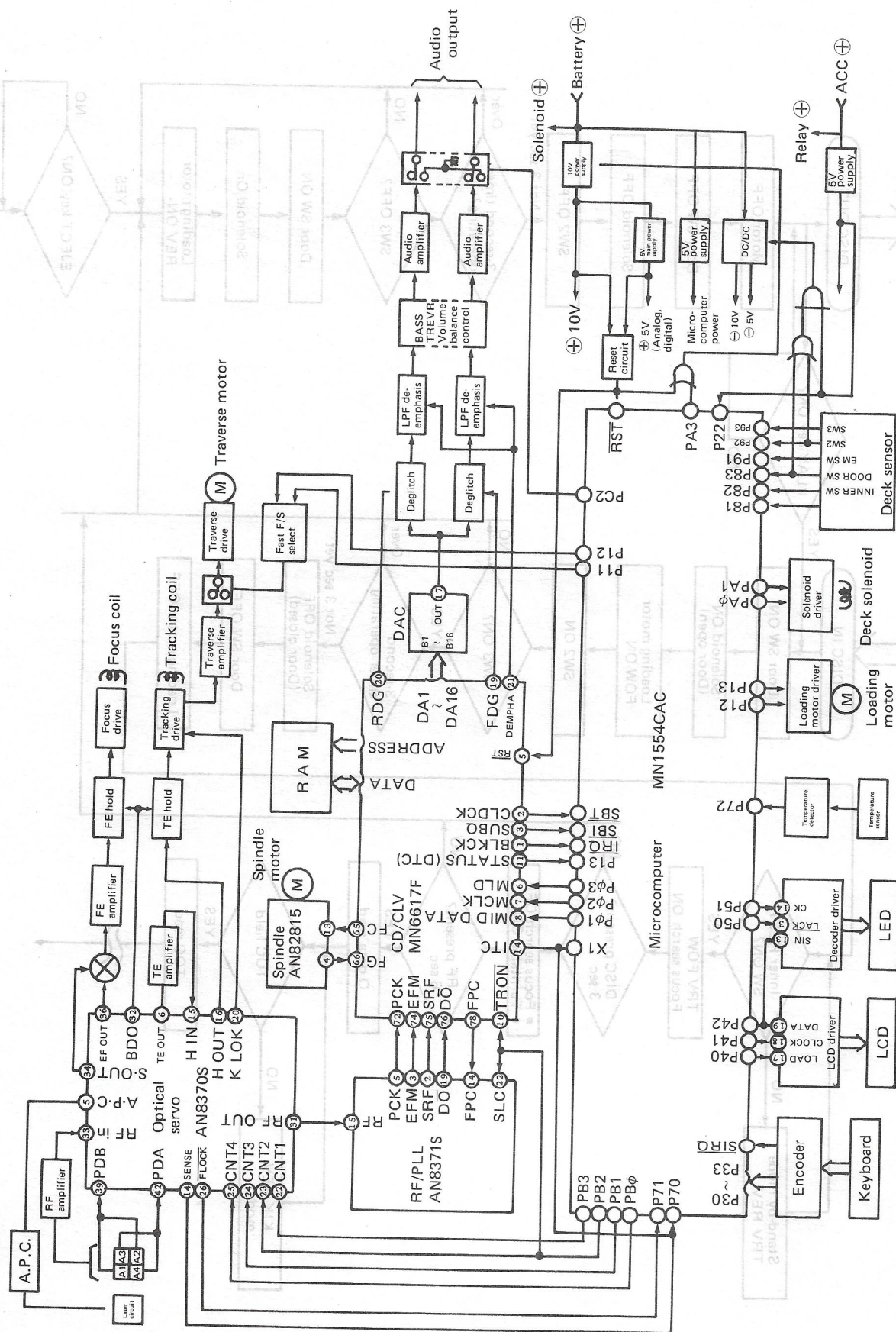
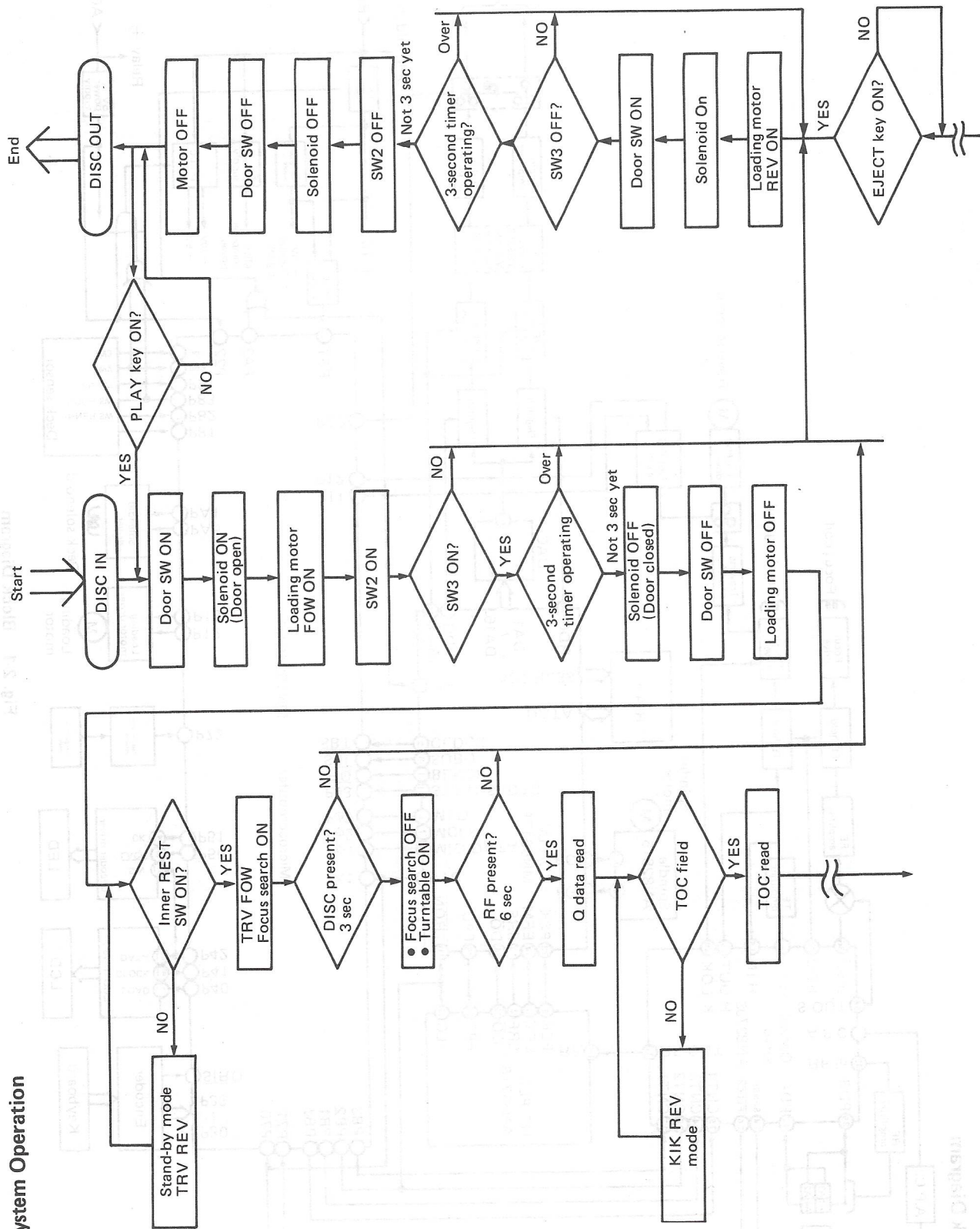


Fig. 2-1 Block Diagram

2-2 CD System Operation



1-3 Tracking Error Generation

Fig. 1-3 shows the basic configuration of the single beam type (phase difference detecting method). The output of the optical pickup is the sum of each diagonal element of the 4-divided photo-detector.

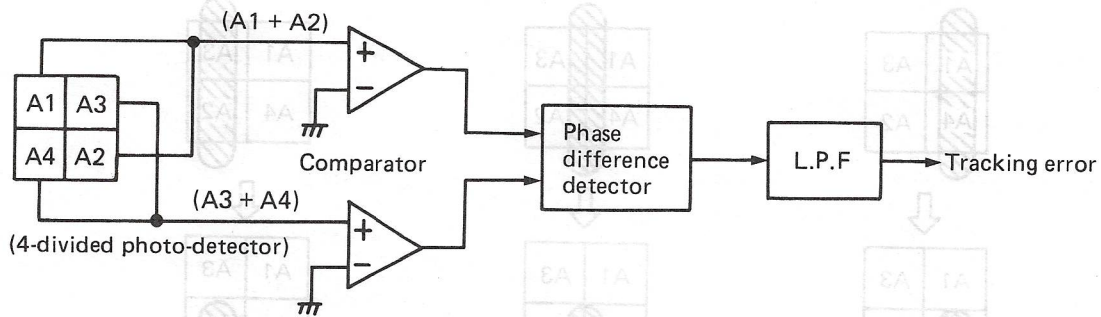
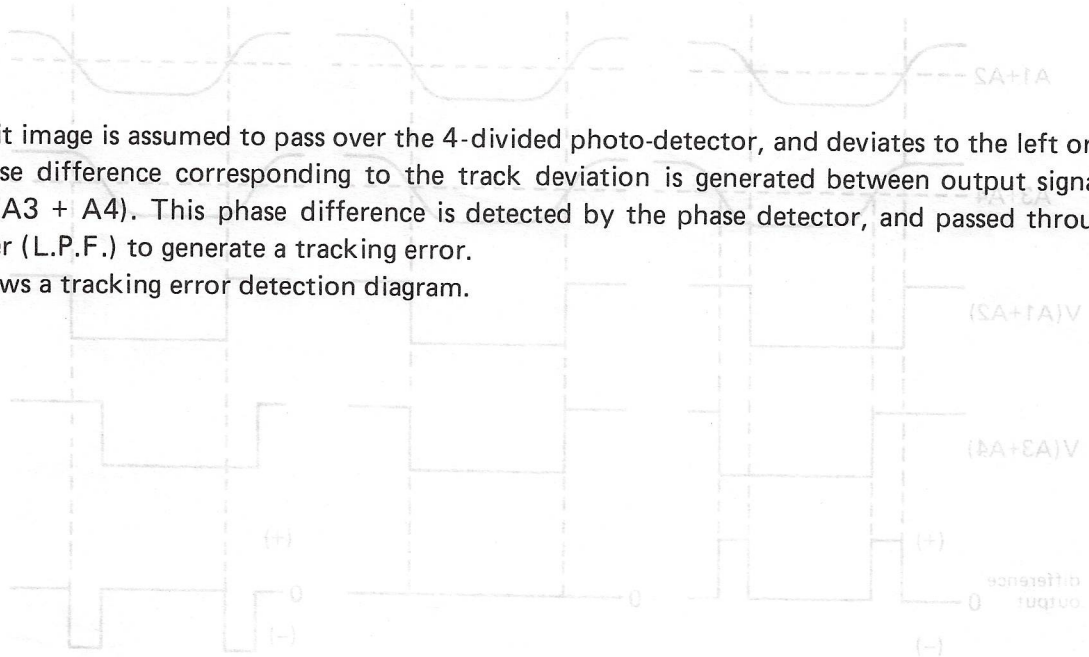


Fig. 1-3 Basic Configuration

If a track pit image is assumed to pass over the 4-divided photo-detector, and deviates to the left or to the right, a phase difference corresponding to the track deviation is generated between output signals $(A1 + A2)$ and $(A3 + A4)$. This phase difference is detected by the phase detector, and passed through the lowpass filter (L.P.F.) to generate a tracking error.

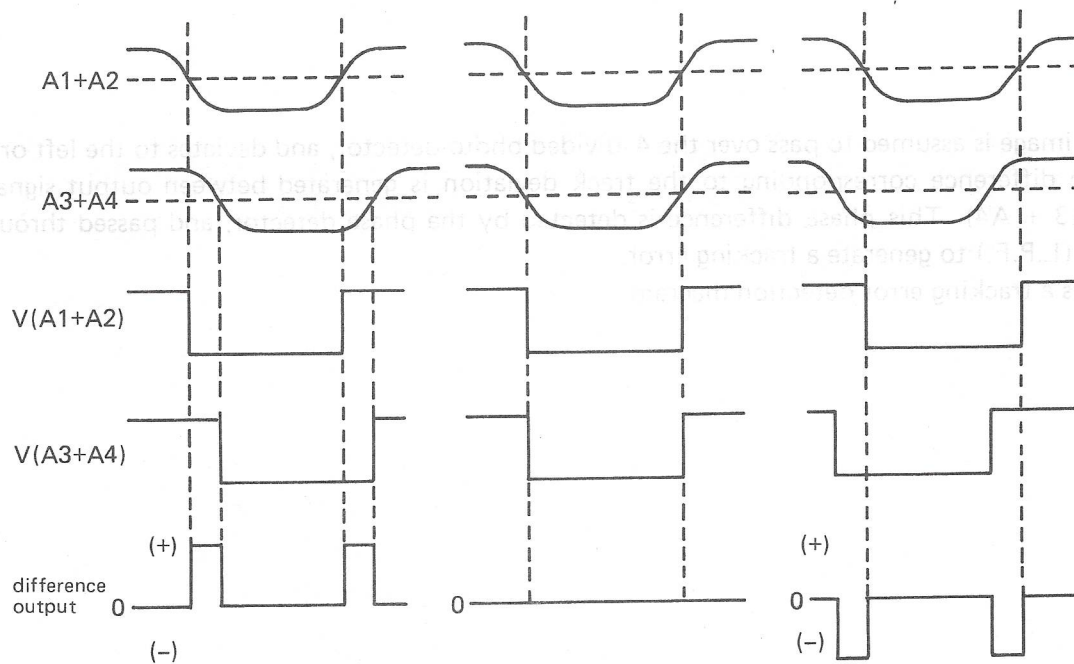
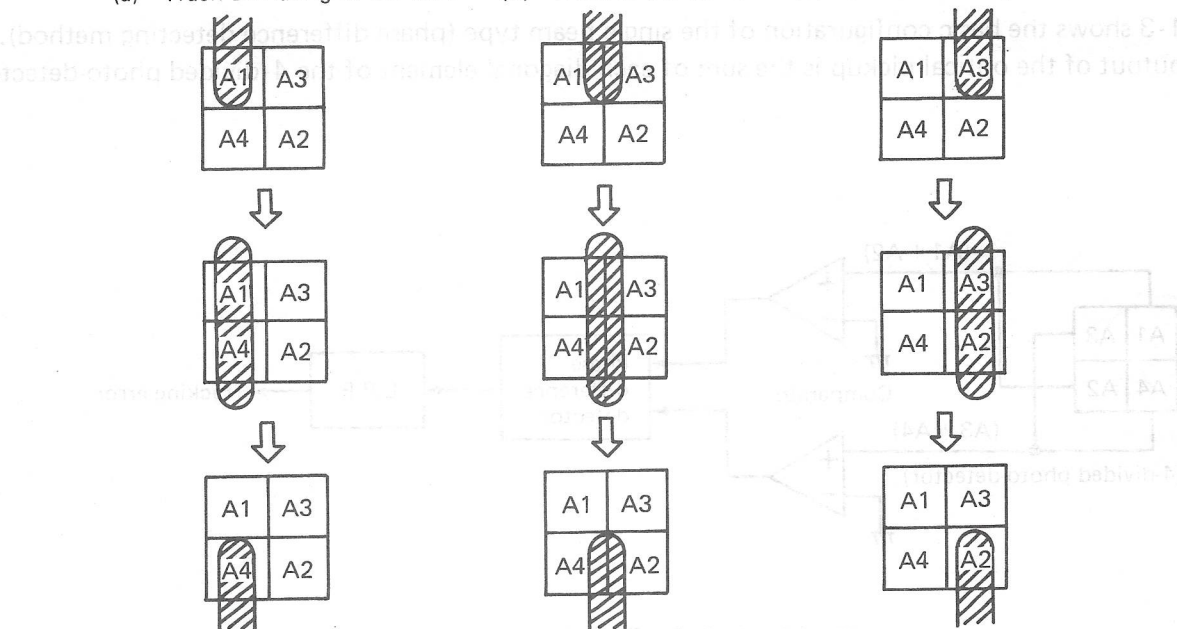
Fig. 1-4 shows a tracking error detection diagram.



• A positive pulse is generated if $V(A1 + A2)$ is higher than $V(A3 + A4)$ or a negative pulse is generated in the opposite case.

Fig. 1-4 Tracking Error Detection Diagram

(a) Track Deviating to the Left (b) Track in the Center (c) Track Deviating to the Right



- A positive pulse is generated if $V(A1 + A2)$ is earlier than $V(A3 + A4)$, or a negative pulse is generated in the opposite case.

Fig. 1-4 Tracking Error Detection Diagram

1. TRAVERSE MECHANISM and OPTICAL PICKUP

1-1 Traverse Mechanism

Fig.1-1 shows the traverse mechanism.

The traverse mechanism conveys power from the motor to the wormgear, gear A, gear B, and main gear, and drives the optical pickup with the rack gear.

Gear A and gear B are double clutch gears.

The inner REST switch is closed by the optical pickup base when the optical pickup is returned to the innermost position by the traverse mechanism.

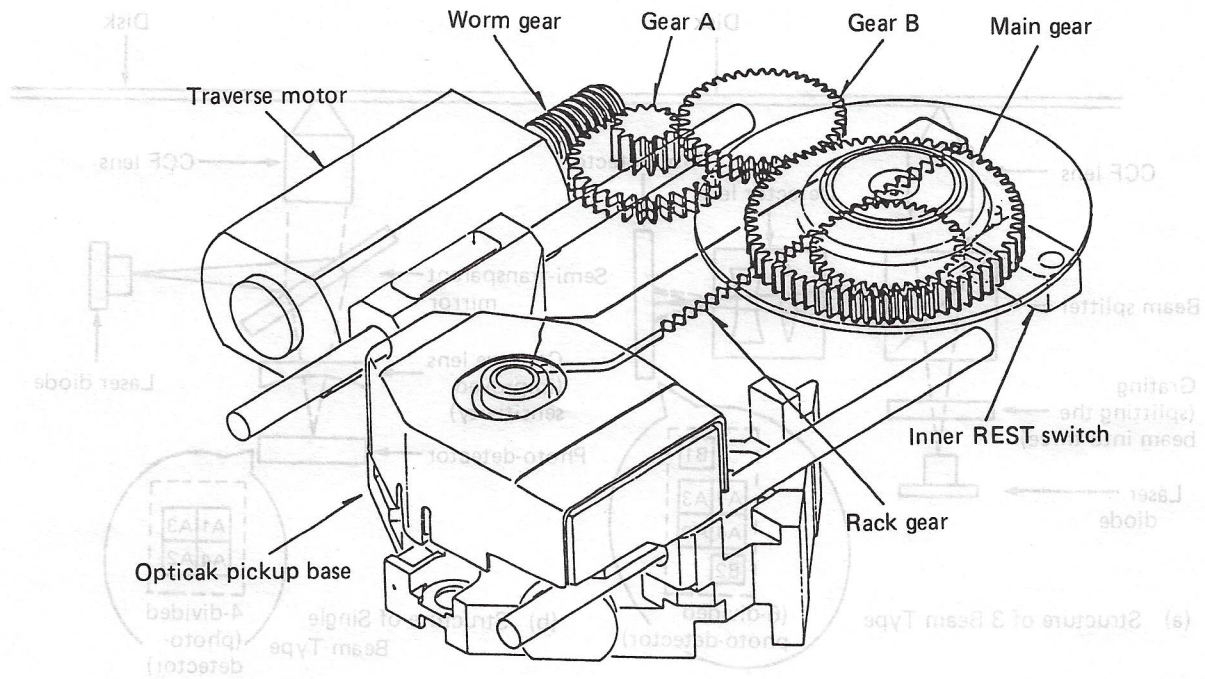


Fig. 1-1 Traverse Mechanism

1-2 3-Beam Type and Single Beam Type

The optical pickup of the SL-XP7 is a single beam type instead of a 3-beam type. The structure of the optical pickup is shown in Fig. 1-2.

The largest difference in optical structure between the 3-beam type and single beam type is that the single beam type has no diffraction grating. Therefore, the single beam type has only beam on the optical path, and its photo-detector is divided into four instead of six (four plus two).

The two sub-beams of the 3-beam type have the function of generating tracking errors. The process of generating tracking errors by the single beam type that is without sub-beams is described in the following.

Focus errors are generated by the astigma method as in conventional models.

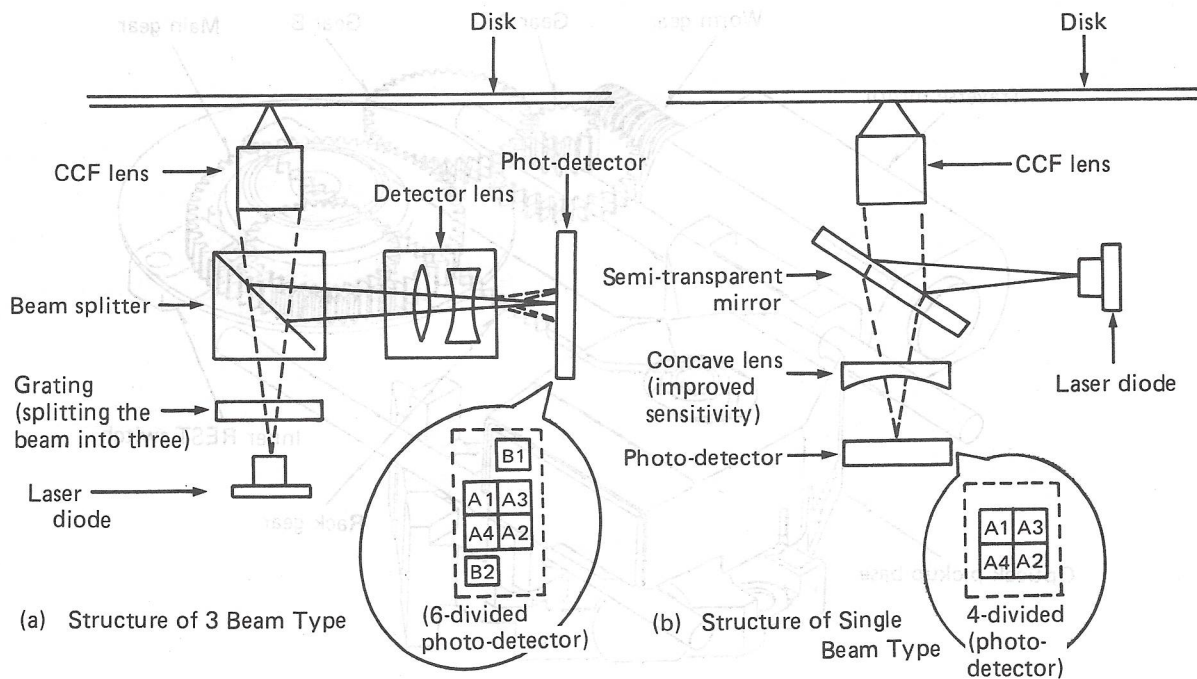


Fig. 1-2 Structure of Optical Pickup

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CIRCUIT OPERATION

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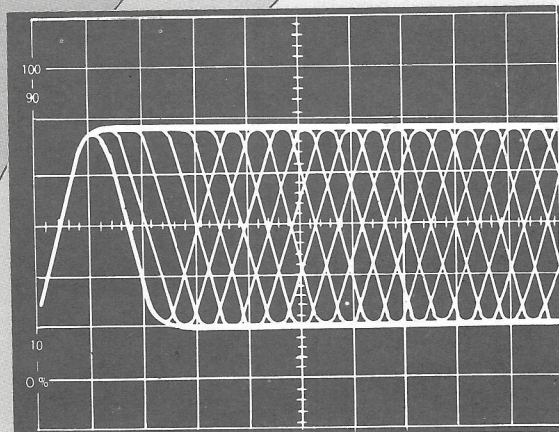
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